

A Study into Cognitive Frameworks, Decision-Making Mechanisms, and Neural Specialisation

A comprehensive briefing synthesising research on dual-process cognition, heuristic biases, functional brain specialisation, and the enduring debate over conscious agency with implications for clinical and legal practice.

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EXECUTIVE BRIEFING

COGNITIVE NEUROSCIENCE



Critical Takeaways: 1 of 2

1

Dual-Process Theory

Kahneman's framework posits two cognitive systems: System 1 is fast, automatic, and unconscious, handling most daily activity but prone to bias; System 2 is slow, deliberate, and effortful, engaging for complex reasoning. System 1 is the default, so its outputs are usually accepted unless System 2 actively overrides them.

2

Heuristics and Anchoring Effect

Tversky and Kahneman showed that an initial value can systematically skew later estimates, even when the anchor is arbitrary. Related heuristics compound this: availability inflates vivid events, representativeness produces the conjunction fallacy, framing changes choices, and sunk costs sustain failing endeavours.

3

Neural Modularity

Discrete cortical regions serve specific functions: the FFA recognises faces, V5/MT detects motion, V4 processes colour, the PPA encodes spatial environments, and the prefrontal cortex governs executive control. Split-brain research confirmed hemispheric specialisation. Left for language and analysis, Right for spatial and holistic processing.

4

Freud and the Unconscious

Freud's structural model divides the psyche into Id, Ego, and Superego. The unconscious shapes behaviour through repression, displacement, and projection. Libet's EEG experiments showed readiness potential precedes conscious intention by about 350ms, suggesting voluntary actions begin unconsciously.

Critical Takeaways: 2 of 2

1

Jung and the Collective Unconscious

Jung proposed a collective unconscious shared across humanity, populated by archetypes such as the Persona, Shadow, Anima/Animus, and Self. Individuation integrates these elements into awareness and represents psychological maturity. He also introduced introversion/extraversion and the complex.

2

Freud vs. Jung: The Great Divide

Their collaboration ended in rupture over deep theoretical disagreements: Freud treated libido as sexual and the Oedipus complex as universal; Jung reframed libido as general psychic energy and rejected sexual reductionism. Freud's model was mechanistic, while Jung's was teleological and meaning-centred.

3

Replicability Crisis

Social priming studies, including Bargh's elderly priming work, failed to replicate in multiple independent attempts. Analysis found 11 of 12 key studies had an R-Index below 50%, pointing to inflation from small samples and publication bias. The crisis drove reforms such as pre-registration, larger samples, open data, and adversarial collaboration.

4

Clinical and Legal Implications

The M'Naghten Rules exempt defendants where a disease of the mind prevents knowledge of the act or its wrongness, mapping directly onto System 2 failure. Diminished responsibility can reduce murder to manslaughter, and syndromes like Anton Syndrome, Riddoch Syndrome, and prosopagnosia show how modular perception affects capacity, intent, and culpability.